Three Views of a Limit
Answer Key

Precalculus / Introduction to Limits

Quick Answers

1. 6
2. 8
3. 4
4. 25
5. 0.4988, 0.5013, $\frac{1}{2}$
6. 3
7. 2, 3
8. 6
9. $\frac{1}{6}$
10. 2.0505, 2.005, 2
11. 3
12. $\frac{31}{32}$, 1

Common wrong answers

2. *If they got 0: read the indeterminate form 0/0 as the answer 0 instead of rewriting*
4. *If they got 16: stopped one term early, adding only the terms through n = 4*
5. *If they got 2: after rationalizing, reported the denominator sqrt(1)+1 instead of its reciprocal*
6. *If they got 2: dropped the middle term and factored the cube as (x-2)(x^2+4), giving 8/4*
8. *If they got 3: took the n = 1 term as the first term, using a = 3/2 in a/(1-r)*
10. *If they got 0.5: divided the leading coefficients the wrong way round, reporting 1/2*
11. *If they got 0.5: read the y-intercept off the graph instead of the end behaviour, giving 1/2*
12. *If they got 2: used a = 1 in a/(1-r) instead of the actual first term a = 1/2*

1. Numerical — read the trend, both sides.

From the left: 5.9, 5.99, 5.999. **From the right:** 6.1, 6.01, 6.001. Both columns close in on the same number, so the limit is that number. $\boxed{6}$

Expected sentence: the limit describes where f is heading as x approaches 3, and the definition deliberately never evaluates f at 3 itself. Since $f(x) = x + 3$ for every $x \neq 3$, the approach is perfectly ordinary even though $f(3)$ is undefined — a hole in the graph, not a break in the trend.

2. Algebraic — factor and cancel.

Step 1. Substitution gives $\frac{0}{0}$, an indeterminate form, so rewrite rather than conclude.

Step 2. Difference of squares: $\dfrac{x^2 - 16}{x - 4} = \dfrac{(x-4)(x+4)}{x-4} = x + 4$ for $x \neq 4$.

Step 3. Now substitution is legal: $4 + 4$. $\boxed{8}$

Common wrong answer: 0 — reading $\frac{0}{0}$ as a value. It is not one; it is the signal that a common factor is hiding in both parts.

3. Graphical — a hole does not stop a limit.

Read the graph. As x moves toward 2 from the left the curve rises toward height 4; from the right it falls toward the same height 4. Both sides agree. $\boxed{4}$

Expected sentence: the open circle records that $g(2)$ is undefined, which is a statement about the single point $x = 2$. The limit is about the values of g at inputs *near* 2, and those are unaffected. Algebraically $g(x) = x + 2$ for $x \neq 2$, which is why the graph is a straight line with one point removed.

4. Series — a finite partial sum.

Expand the sigma notation by substituting $n = 1, 2, 3, 4, 5$:

$$\sum_{n=1}^{5}(2n - 1) = 1 + 3 + 5 + 7 + 9$$

Add: $1 + 3 = 4$, $+5 = 9$, $+7 = 16$, $+9 = 25$. $\boxed{25}$

Worth noticing: the running totals are $1, 4, 9, 16, 25$ — the perfect squares. The kth partial sum of the odd numbers is k^2.

Common wrong answer: 16 — the sum stopped at $n = 4$. The upper limit of the sigma is inclusive, so $n = 5$ contributes a term too.

5. Numerical — build the table, then name the exact value.

Left entry ($x = -0.01$): $\dfrac{\sqrt{0.99} - 1}{-0.01} = \dfrac{-0.0050126\ldots}{-0.01} = 0.5012562\ldots$ $\boxed{0.5013}$

Right entry ($x = 0.01$): $\dfrac{\sqrt{1.01} - 1}{0.01} = \dfrac{0.0049876\ldots}{0.01} = 0.4987562\ldots$ $\boxed{0.4988}$

The exact limit. The two entries squeeze in on 0.5 from opposite sides. Confirm it algebraically with the conjugate: $\dfrac{\sqrt{x+1} - 1}{x} \cdot \dfrac{\sqrt{x+1} + 1}{\sqrt{x+1} + 1} = \dfrac{x}{x(\sqrt{x+1} + 1)} = \dfrac{1}{\sqrt{x+1} + 1}$, which at $x = 0$ is $\dfrac{1}{1+1}$. $\boxed{\tfrac{1}{2}}$

Common wrong answer: 2 — the denominator $\sqrt{1}+1$ reported instead of the whole fraction. The table rules it out immediately: the entries sit near 0.5, not near 2.

Grading note: the table alone is *evidence*, not proof — it can never rule out something happening between the sampled points. The algebra is what settles it.

6. Algebraic — cubes over squares.

Step 1. Substitution gives $\frac{0}{0}$; both top and bottom vanish at $x = 2$, so both carry a factor of $(x - 2)$.

Step 2. Factor each:

$$\frac{x^3 - 8}{x^2 - 4} = \frac{(x-2)(x^2 + 2x + 4)}{(x-2)(x+2)} = \frac{x^2 + 2x + 4}{x + 2} \quad (x \neq 2)$$

Step 3. Substitute $x = 2$: $\dfrac{4 + 4 + 4}{4} = \dfrac{12}{4}$. $\boxed{3}$

Common wrong answer: 2 — from factoring the cube as $(x-2)(x^2+4)$, dropping the middle term $2x$. The cube pattern is $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$; the middle term is not optional.

7. Graphical — one-sided limits and a jump.

(a) From the left the graph follows the line through $(0,1)$ rising to the open circle at height 2: $\boxed{2}$

(b) From the right the graph starts at the solid dot at height 3 and falls away: $\boxed{3}$

(c) Manual review — expected justification. The two-sided limit exists only when both one-sided limits exist *and* are equal. Here they are 2 and 3, which are not equal, so $\lim_{x\to1} h(x)$ does **not** exist. Accept any answer that compares the two one-sided values; do not accept "because the graph breaks" with no reference to them. Note also that $h(1) = 3$ exists — a function can have a value at a point where the limit does not, which is the mirror image of problem 3.

8. Series — an infinite geometric sum.

Identify the pieces. First term $a = 3$ (take $n = 0$), common ratio $r = \frac{1}{2}$, and $|r| < 1$, so the series converges.

Apply the formula $S = \dfrac{a}{1-r}$: $S = \dfrac{3}{1-\frac{1}{2}} = \dfrac{3}{\frac{1}{2}}$. $\boxed{6}$

The partial-sum limit it represents. $S_k = \displaystyle\sum_{n=0}^{k} 3\left(\tfrac{1}{2}\right)^n = 6\left(1-\left(\tfrac{1}{2}\right)^{k+1}\right)$, and the value above is $\lim_{k\to\infty} S_k$. The sum of an infinite series is defined as that limit.

Common wrong answer: 3 — from taking the $n = 1$ term, $\frac{3}{2}$, as the first term. The index starts at $n = 0$, so $a = 3$.

9. Algebraic — rationalize the numerator.

Step 1. Substitution gives $\frac{0}{0}$, and the trouble is a radical, so multiply by the conjugate of the numerator.

Step 2.

$$\frac{\sqrt{x+9}-3}{x}\cdot\frac{\sqrt{x+9}+3}{\sqrt{x+9}+3} = \frac{(x+9)-9}{x(\sqrt{x+9}+3)} = \frac{1}{\sqrt{x+9}+3} \qquad (x \neq 0)$$

Step 3. Substitute $x = 0$: $\dfrac{1}{3+3}$. $\boxed{\frac{1}{6}}$

10. Numerical — end behaviour from a table.

At $x = 100$: $\dfrac{2(100)+3}{100-1} = \dfrac{203}{99} = 2.0505050\ldots$ $\boxed{2.0505}$

At $x = 1000$: $\dfrac{2(1000)+3}{1000-1} = \dfrac{2003}{999} = 2.0050050\ldots$ $\boxed{2.0050}$

The limit. The values are marching down toward 2. Algebraically, divide top and bottom by x: $\dfrac{2+3/x}{1-1/x}$, and both $3/x$ and $1/x$ go to 0 as x grows. $\boxed{2}$

Common wrong answer: 0.5 — the leading coefficients divided the wrong way round. The table is the check: its entries sit just above 2.

11. Graphical then algebraic — a horizontal asymptote.

Read the graph. Far out in either direction the curve flattens against the dashed line, so the limit is the height of that line. $\boxed{3}$

Confirm algebraically. Divide numerator and denominator by x^2:

$$\frac{3x^2+1}{x^2+2} = \frac{3+\dfrac{1}{x^2}}{1+\dfrac{2}{x^2}} \longrightarrow \frac{3+0}{1+0}$$

Common wrong answer: 0.5 — the height where the curve crosses the y-axis. That is $p(0)$, a single point, not the end behaviour.

Teaching note: a horizontal asymptote *is* a limit statement about end behaviour. The curve may cross an asymptote elsewhere; what matters is where it settles.

12. Synthesis — the sum of a series is a limit.

(a) The fifth partial sum. $S_5 = \frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}$. Over the common denominator 32 this is $\frac{16+8+4+2+1}{32}$. $\boxed{\frac{31}{32}}$

(b) The infinite sum. Geometric with $a=\frac{1}{2}$ and $r=\frac{1}{2}$, so $S = \dfrac{1/2}{1-1/2}$. $\boxed{1}$

Common wrong answer to (b): 2 — from using $a=1$ in $\frac{a}{1-r}$. The first term here is the $n=1$ term, $\frac{1}{2}$.

(c) Manual review — expected explanation. The running totals are $\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \frac{31}{32}$: each one closes half of the remaining gap to 1, so $S_k = 1-\left(\frac{1}{2}\right)^k$ and no partial sum ever reaches 1. Numerically the totals approach 1; graphically the plotted partial sums flatten against the line $y=1$; algebraically $\lim_{k\to\infty} S_k = 1$. The sum of an infinite series is *defined* to be that limit — not the result of an endless addition, which is why an infinite sum can be a finite number at all. Accept any answer that names the sum as a limit of partial sums; do not accept “you keep adding forever”.